

Smartino (صديقي الذكي)

An Intelligent Educational Platform for Egyptian Children

Bridging the gap between Standard Arabic and Egyptian Colloquial dialect through fine-tuned AI.

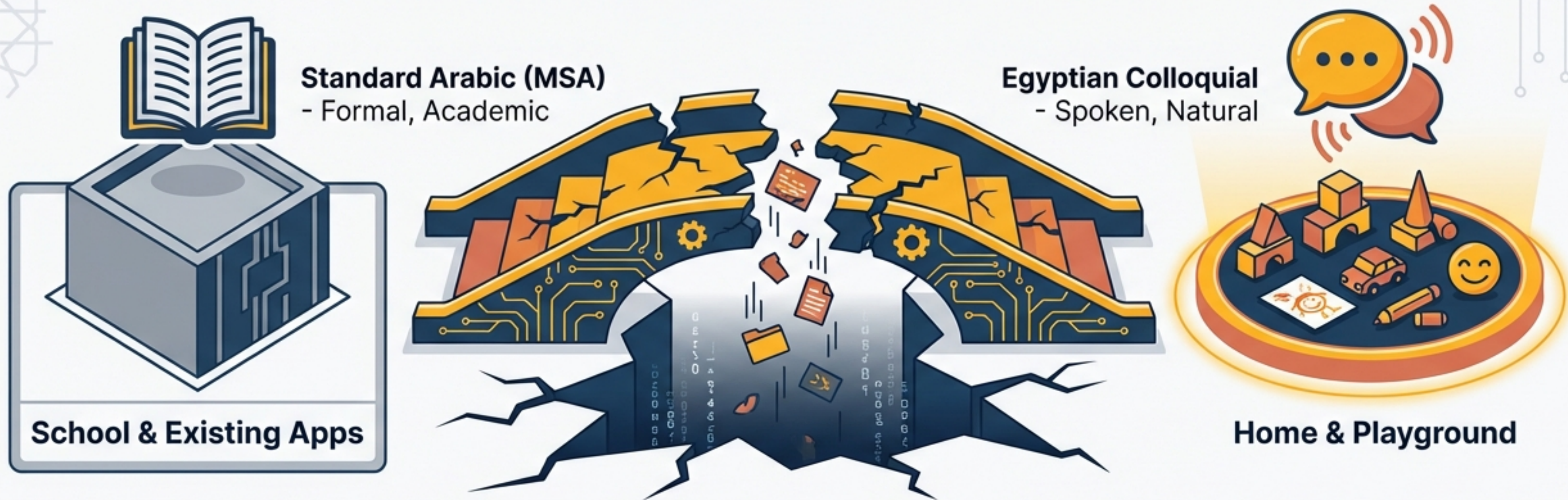
Graduation Project: Department of Computer Science & Information Technology
Academic Year: 2025-2026

Developed by:
[Student 1 Name]
[Student 2 Name]
[Student 3 Name]
[Student 4 Name]

Supervised by: [Supervisor Name]



The Linguistic Gap in Early Childhood Education

**Dialect Mismatch:**

Apps use MSA, alienating children who speak colloquial.

**Engagement Deficit:**

73% of children prefer game-based learning over passive methods.

**Market Void:**

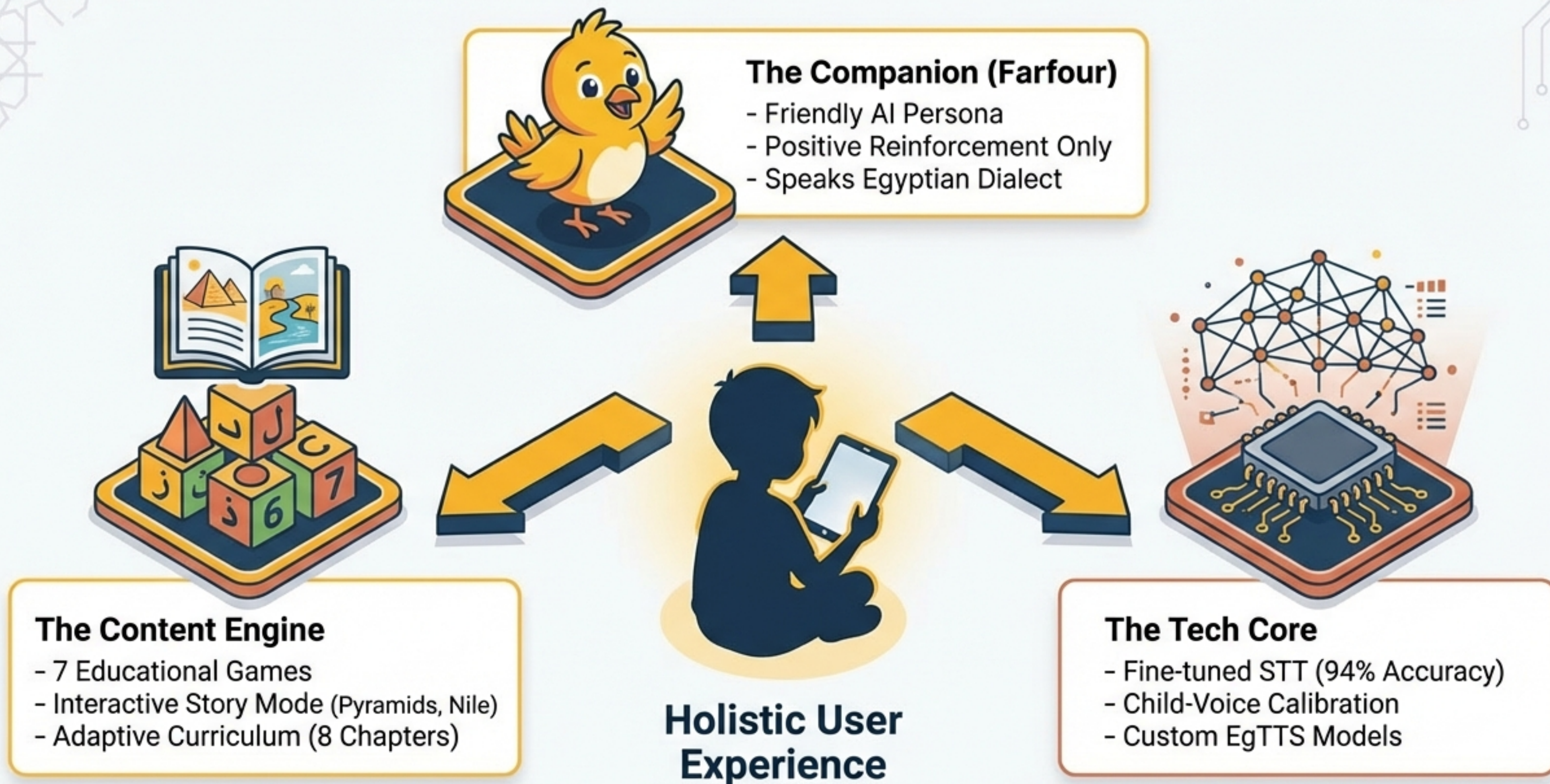
68% of parents struggle to find culturally relevant Arabic apps.

**Parental Anxiety:**

Lack of oversight on AI interactions and safety.

Conclusion: Current tools fail because they ignore the sociolinguistic reality of the Egyptian child.

A Comprehensive AI-Powered Learning Ecosystem



Solving the Dialect Challenge with LoRA Fine-Tuning

Transforming General Purpose AI into a Dialect-Specific Expert

The Fine-Tuning Recipe

Base Model



LoRA
Injection



Smartino Model



Smartino Model

Fine-Tuned
Egyptian Model

76% Baseline
Accuracy
76% 

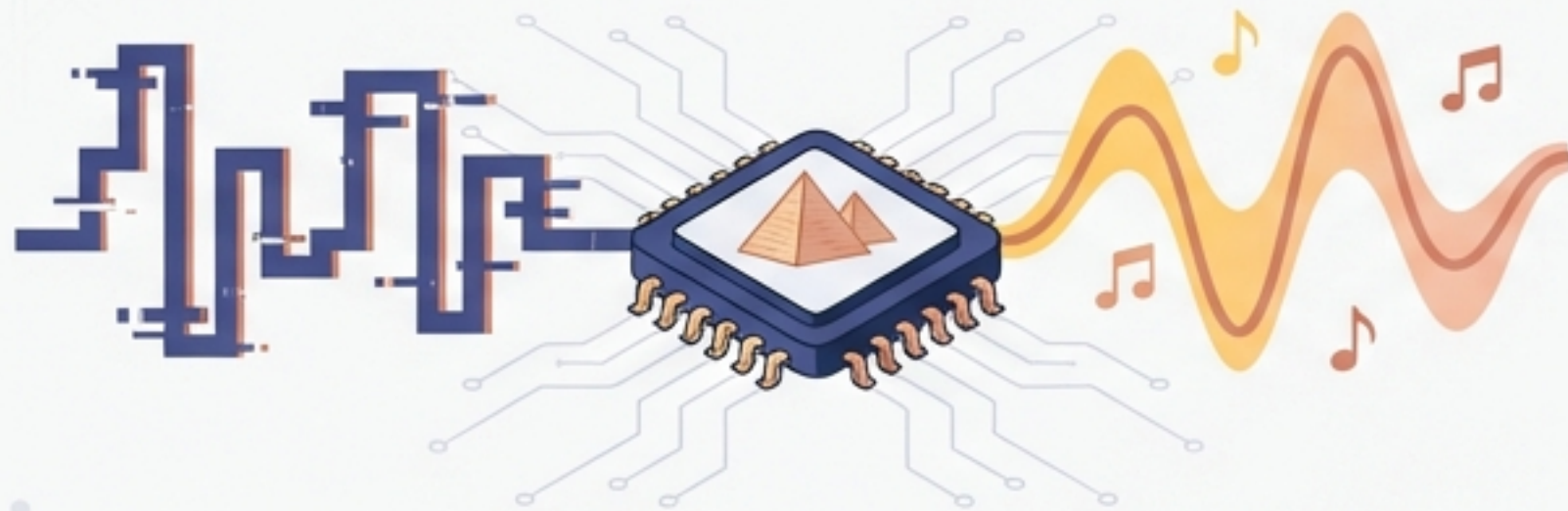
94% 
Accuracy
(+18% Gain)



✓ Verified Dialect Coverage ✓

Engineering Natural Interactions and Child Safety

Generative Voice (EgTTS)



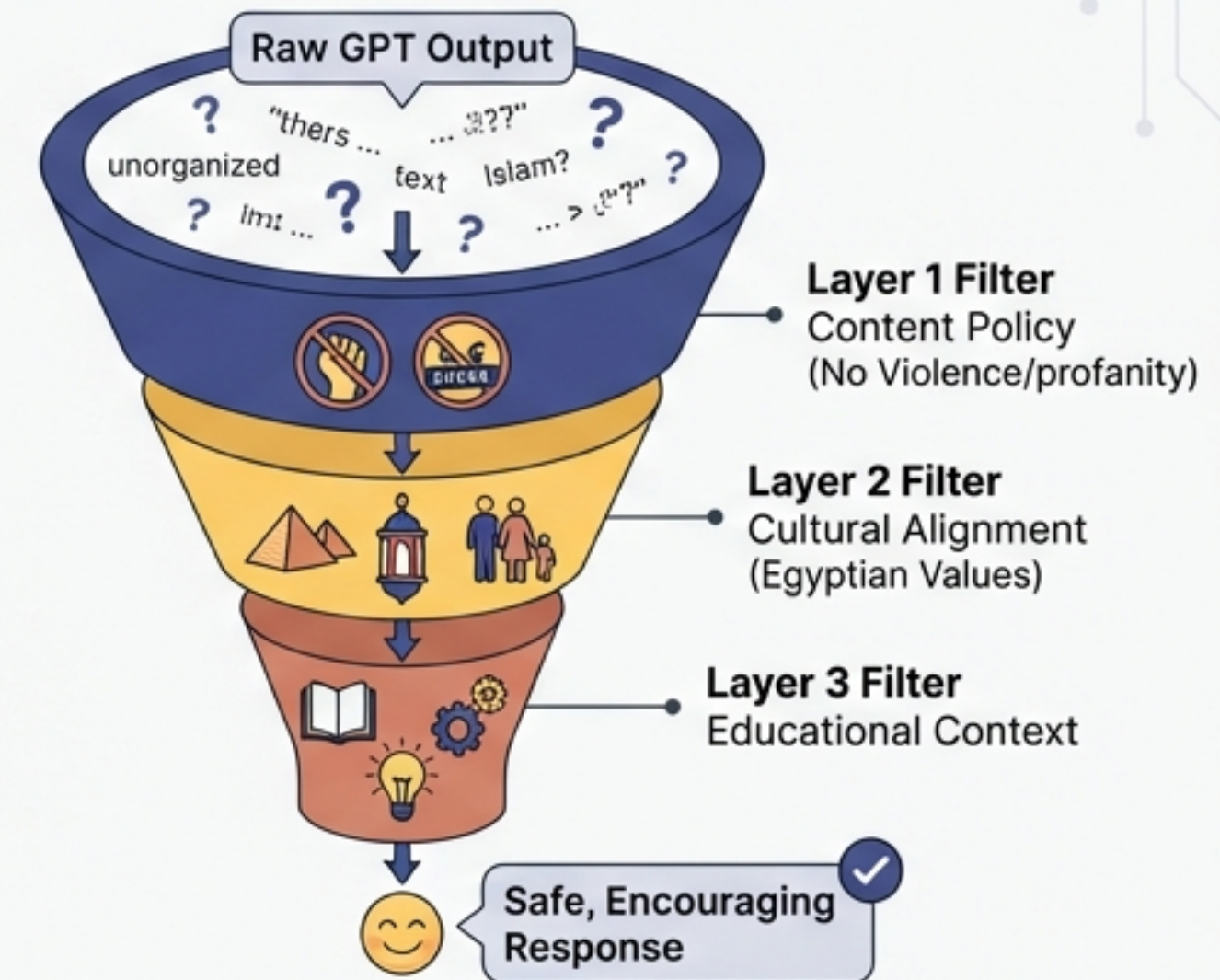
Architecture: Custom VITS + LoRA

Input: 2 hours of reference recordings

Tuning: Prosody adjustment for Egyptian intonation

Result: 4.2/5.0 Mean Opinion Score (MOS)

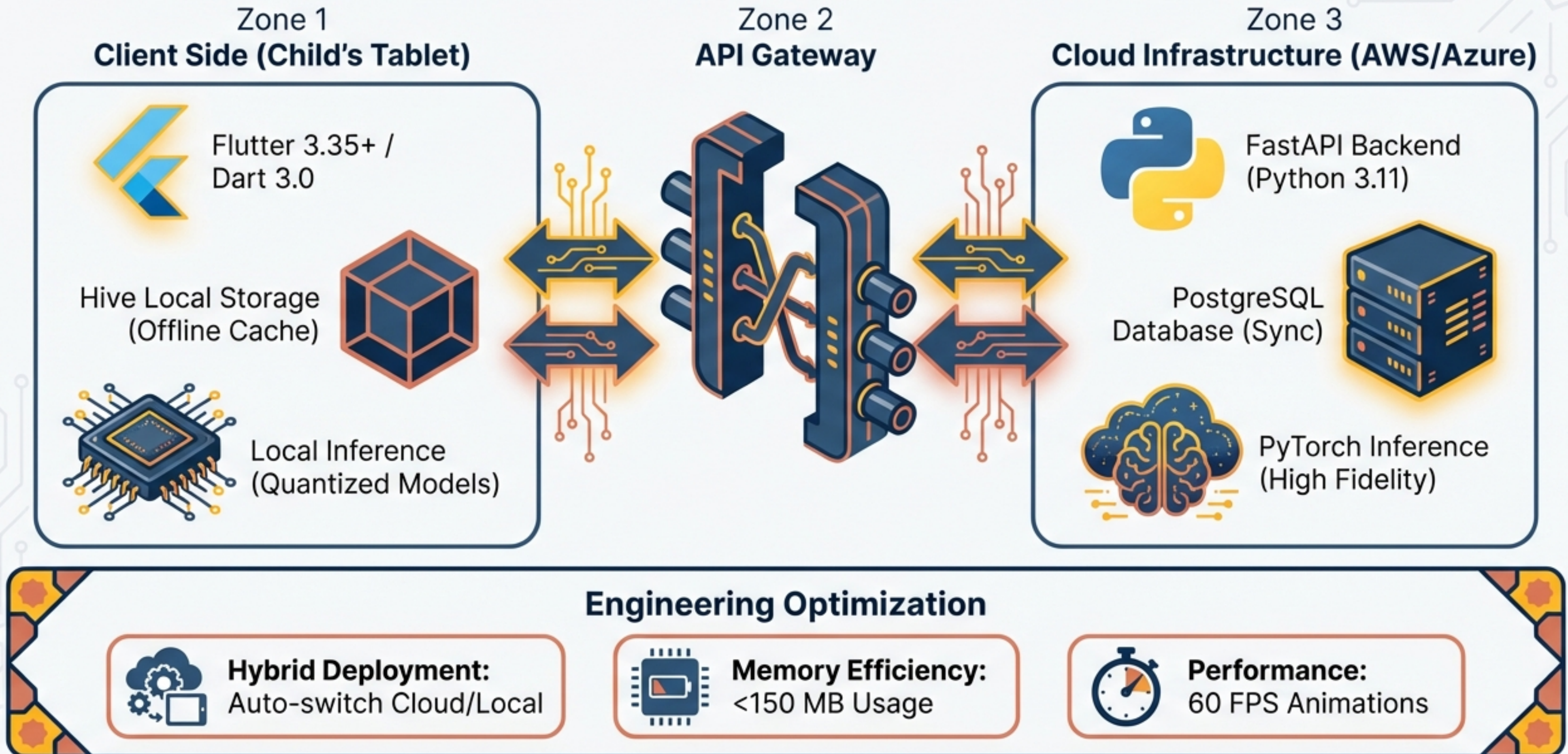
LLM & Safety Protocols



Training: 10K conversations, 500 stories

Safety Rate: 99.9% verified

Orchestrating a Scalable, Hybrid Architecture



Immersive Learning Through Play

Arabic Letter Adventure



Letter Balloons



Missing Letter



Reading Game



Puzzle Challenge



Engagement Mechanics

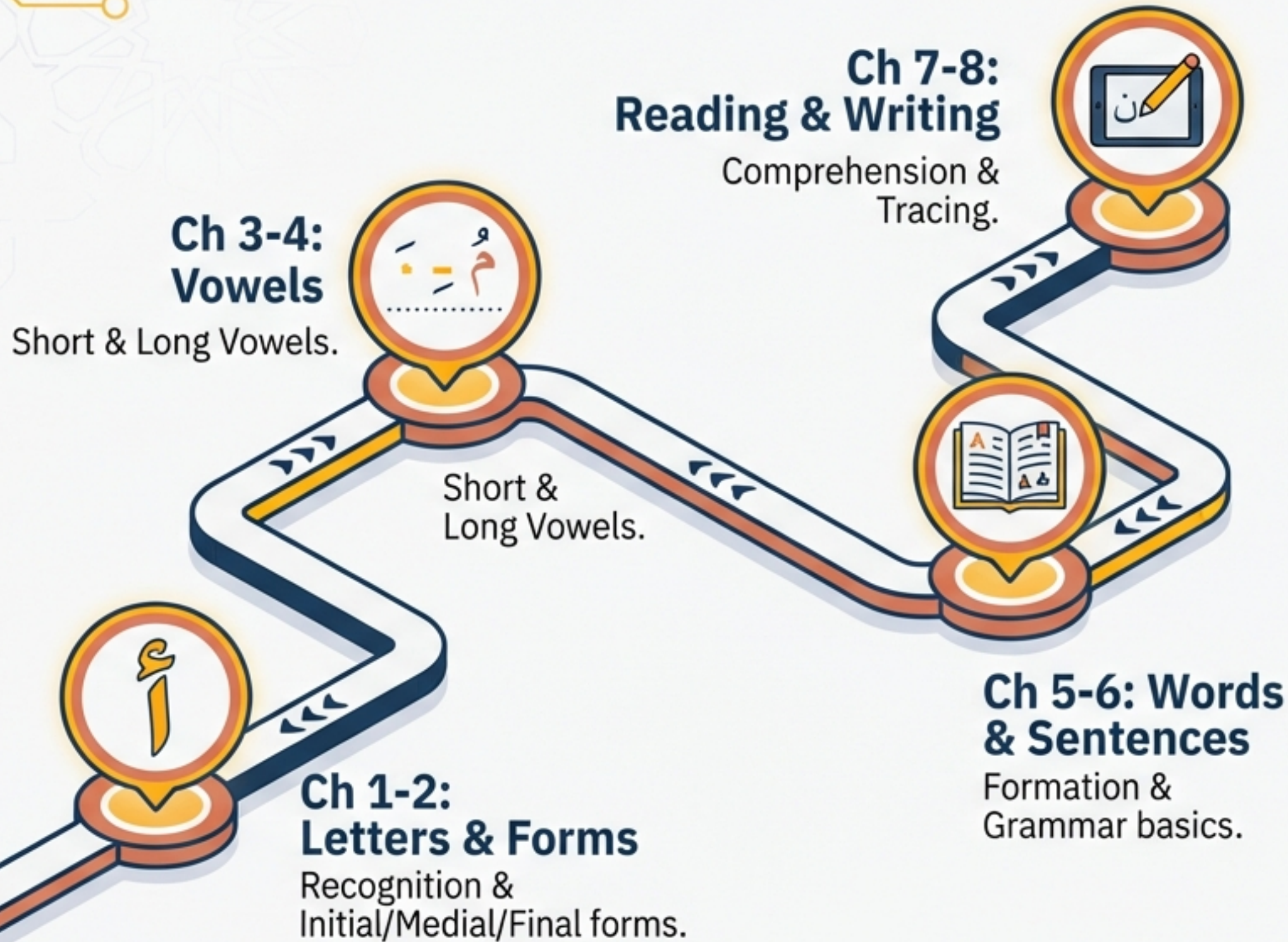
 **Dynamic Rewards:**
1-3 Star rating system

• **Companion Integration:**
'Farfour' appears for praise



 **Adaptive Difficulty:**
Scales with player speed and accuracy

A Structured 8-Chapter Pedagogical Curriculum



Scope & Progression

- **20+** Interactive Stages
- **100+** Unique Activities
- **Mastery Gate:** Stages unlock via ★★ ★ accumulation.

Empowering Parents with Radical Transparency



The 'Black Box' Reveal



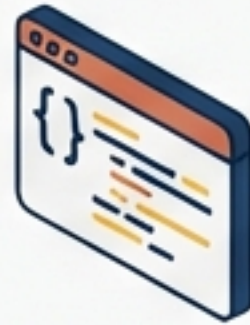
Control Center Features

- **Radical Transparency:** Full access to conversation logs.
- **Safety Monitoring:** Automated flags for flagged topics.
- **Screen Time:** Configurable limits.
- **Curriculum Tracking:** Granular progress reports.

Implementation Metrics & Engineering Rigor

Codebase Scale

18,500+



Lines of Code (Flutter/Python)

Development Velocity

800+



Team Hours over 10 Weeks

Complexity

65+



Files & Components

App Efficiency

<2s



Cold Boot Load Time

Optimization

45 MB



Total App Size

Project Status

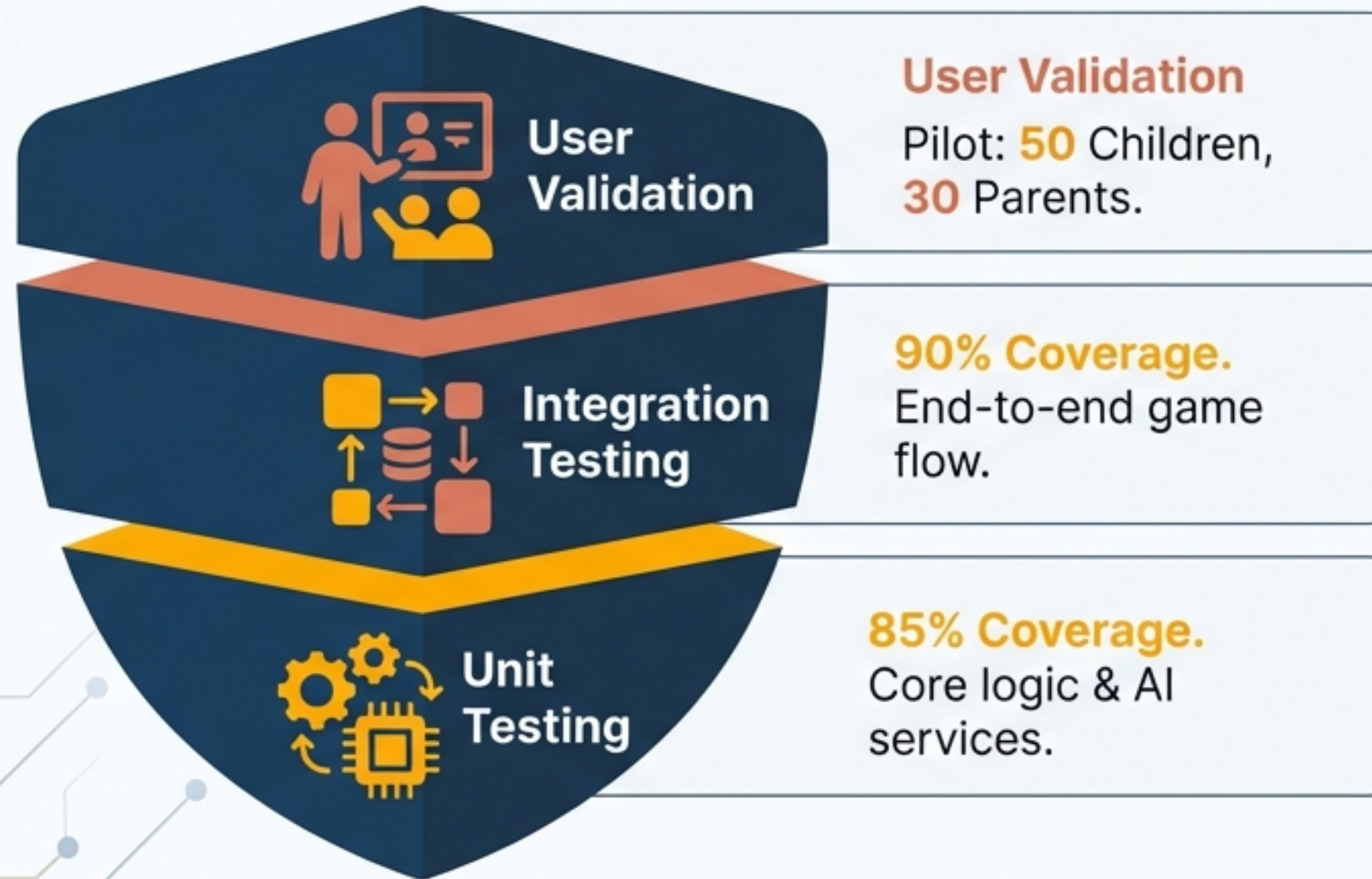
98%



Production Ready / Complete

Cairo

A Comprehensive Multi-Layer Testing Strategy



Security & Accessibility Protocols



Encryption: AES-256 Data Protection.



Auth: OAuth 2.0 Standard.



Compliance: GDPR & Child Safety Standards.

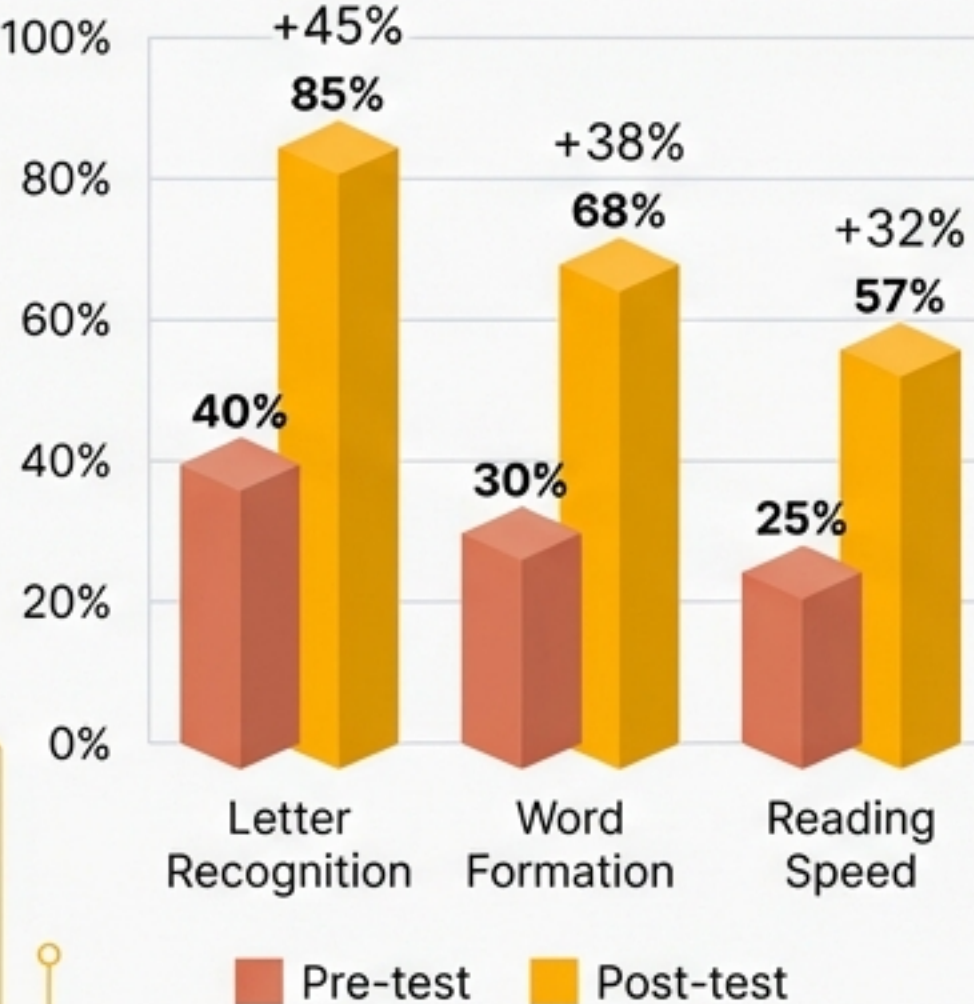


Access: Colorblind modes & Screen reader support.

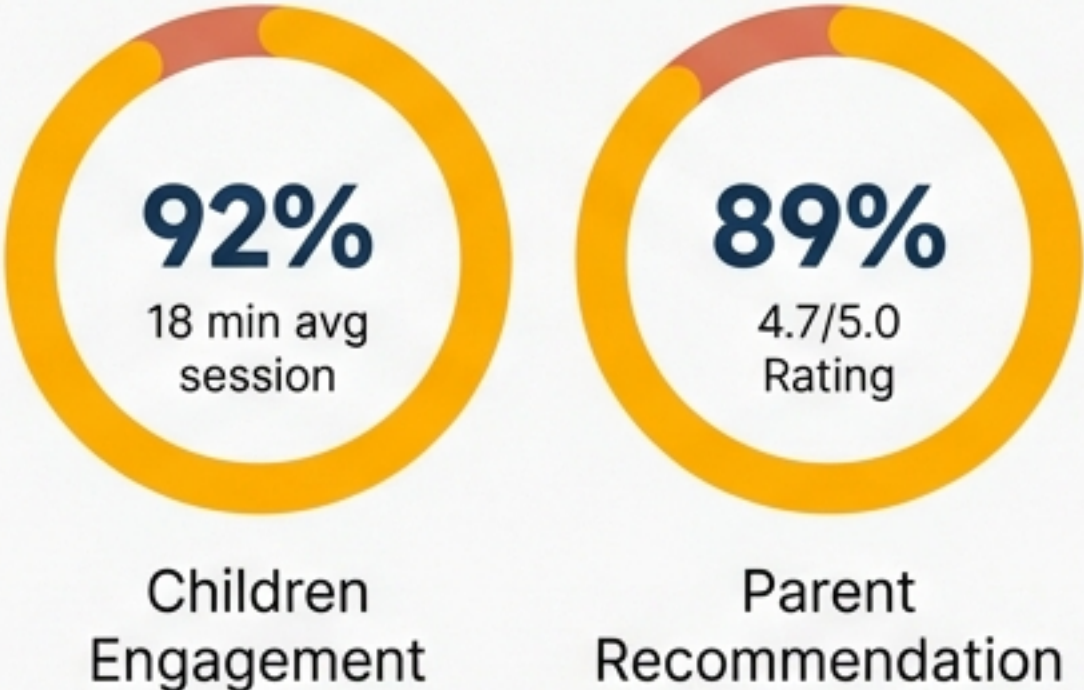
Zero critical bugs found in production environment. 🚀

Measurable Impact and Pilot Results

Learning Outcomes (n=50 Pilot)



User Satisfaction



Technical Targets

	Target	Actual	
STT Accuracy	90%	94%	✓
Response Time	<1s	0.5s	✓

Overcoming Technical & Regional Challenges

Dialect Diversity



Challenge: Multiple regional dialects (Cairo vs Delta).



Solution: Multi-dialect training data sets.



Result: 93%+ accuracy across regions.

Child Voice Variability



Challenge: High pitch & erratic pronunciation.



Solution: Pitch augmentation & age-specific tuning.



Result: Robust detection for ages 5-10.

Offline Functionality



Challenge: Unreliable internet access.



Solution: Quantized local models + Smart Caching.



Result: Full core gameplay offline.

Real-Time Latency



Challenge: AI processing delay.



Solution: Hybrid deployment.



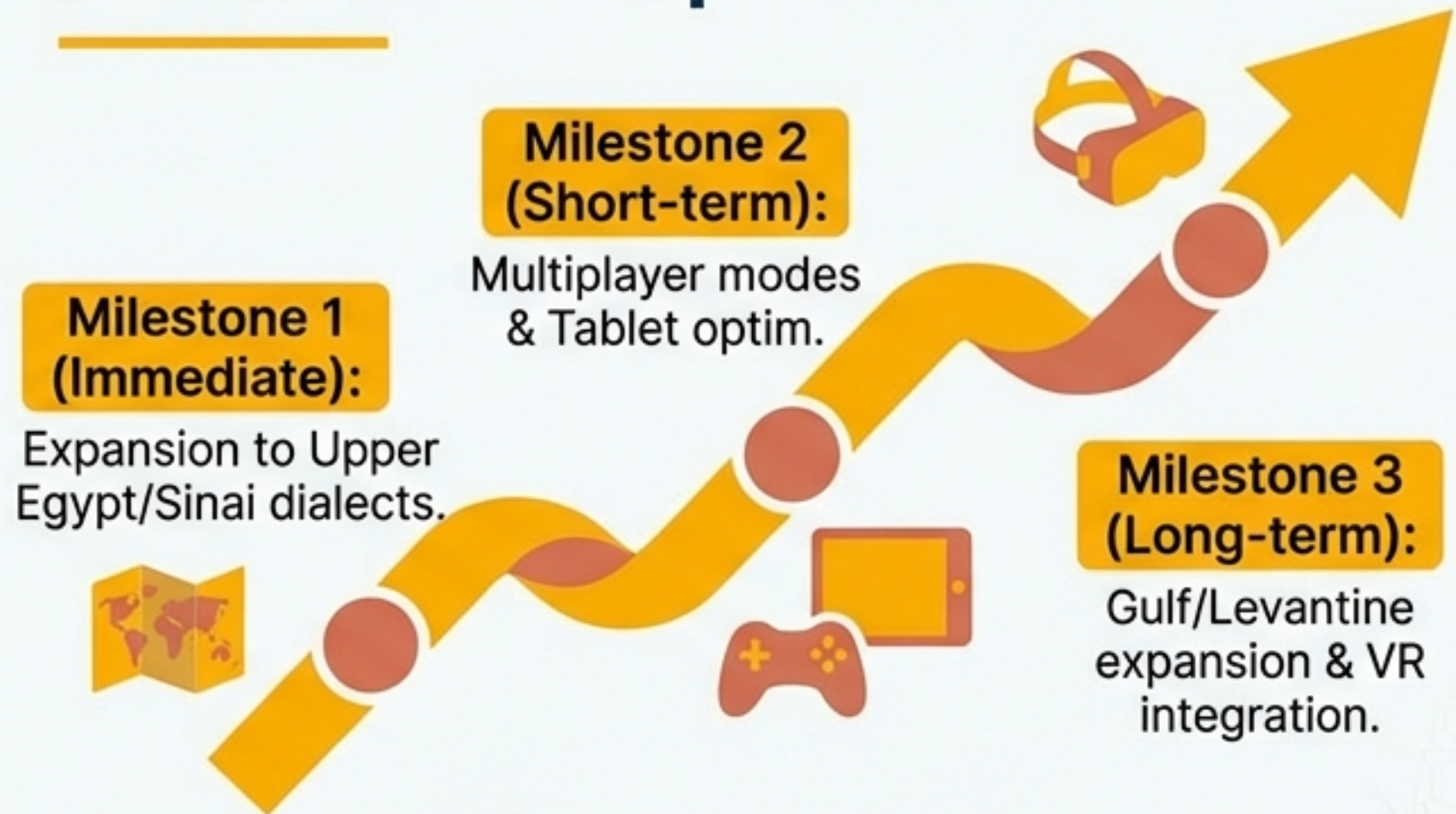
Result: <500ms response time.

Roadmap and Final Conclusions

Novel Contributions

- ✓ First fine-tuned Whisper model for Egyptian children.
- ✓ Hybrid Cloud/Local AI architecture for education.
- ✓ Total transparency framework for Parent-AI trust.

Future Roadmap



Smartino proves that AI can be safe, culturally grounded, and technically brilliant—transforming how Egyptian children connect with their language.